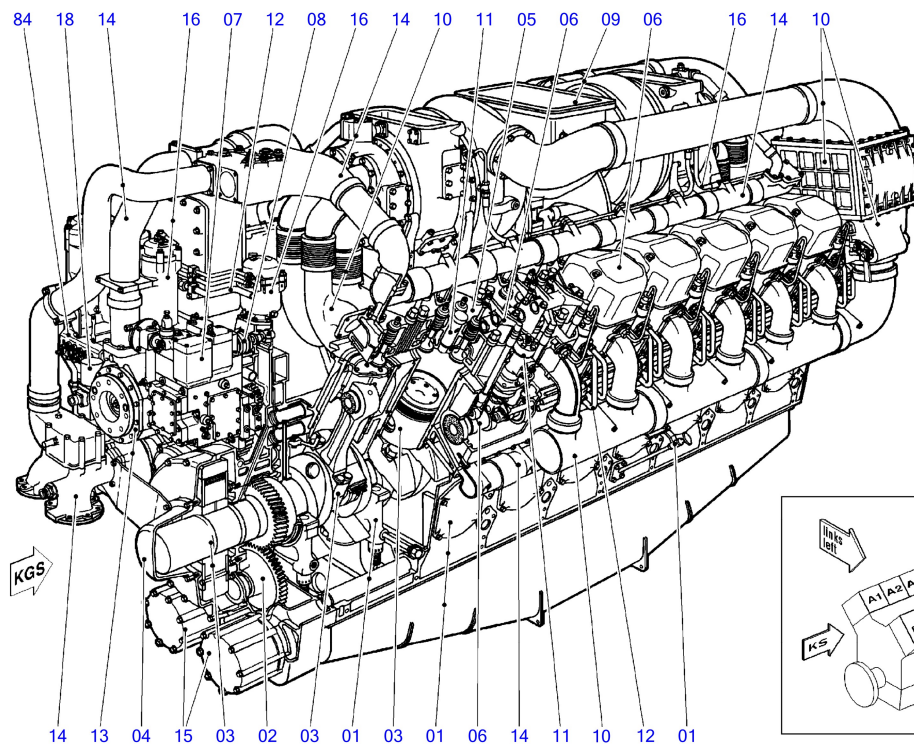




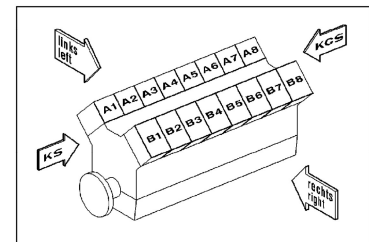
Gruppen - Übersicht

Group Review

16 V 956 TB .2
 16 V 1163 TB .2



Gilt sinngemäß auch für
 Applies analogously to
 12 V 956 TB .2
 12 V 1163 TB .2
 20 V 956 TB .2
 20 V 1163 TB .2



E 000 5723 a0

- | FG | FG DESIGNATION |
|----|-----------------------------------|
| 01 | CRANKCASE |
| 02 | GEAR TRAIN |
| 03 | RUNNING GEAR |
| 04 | COVER PLATES AT KGS |
| 05 | CYLINDER HEAD |
| 06 | CONTROL SYSTEM |
| 07 | ENGINE GOVERNOR |
| 08 | GOVERNOR LINKAGE |
| 09 | EXHAUST TURBOCHARGER |
| 10 | CHARGE AIR PIPING-EXHAUST PIPING |
| 11 | FUEL INJECTION EQUIPMENT |
| 12 | FUEL SYSTEM |
| 13 | COOLANT PUMP |
| 14 | COOLANT SYSTEM |
| 15 | OIL PUMP |
| 16 | OIL SYSTEM |
| 17 | DRIVE FOR SPEED MONITORING |
| 18 | STARTING EQUIPMENT |
| 19 | ENGINE MOUNTING |
| 84 | MONITORING EQUIPMENT |
| 86 | ACCESSORIES (ELECTRIC, PNEUMATIC) |

mtu
Dieselmotor

Motor und Turbinen Union
Frederickshafen GmbH

Hinweise für Betrieb und Wartung

Modell	
Bestellnr.	
Gepl. Lebensdauer h	
Standardleistung kW	
Maximale Leistung kW	
Leistung kW	

Bitte beachten: Vor dem Betrieb des Motors muss die Wartung gemäß den Vorschriften der MTU durchgeführt werden.

Warning

Do not open hot crankcase for at least 10 minutes after shutdown of engine.
Do not restart overheated engine before cause of overheating has been remedied.

Vor dem Öffnen des Motors muss die Wartung gemäß den Vorschriften der MTU durchgeführt werden.

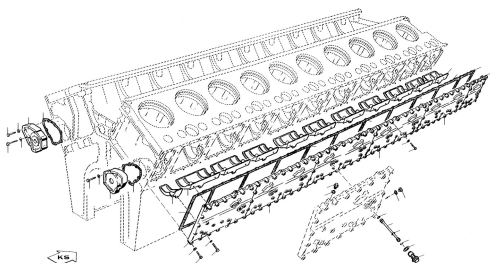
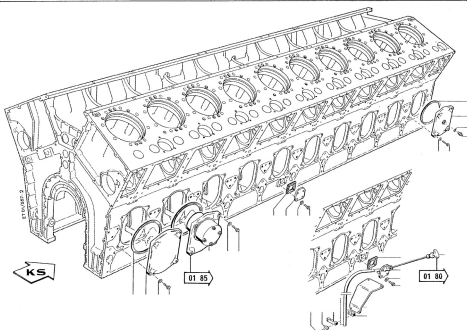
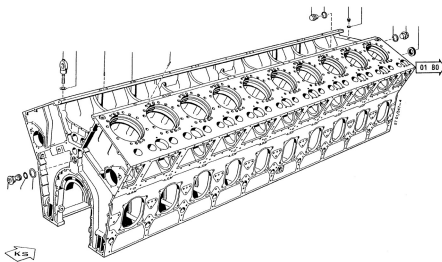
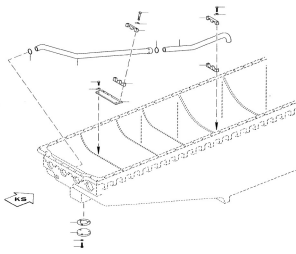
Bitte beachten: Vor dem Betrieb des Motors muss die Wartung gemäß den Vorschriften der MTU durchgeführt werden.

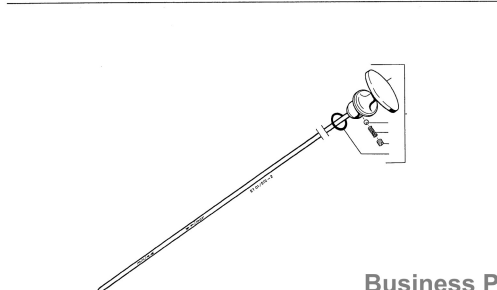
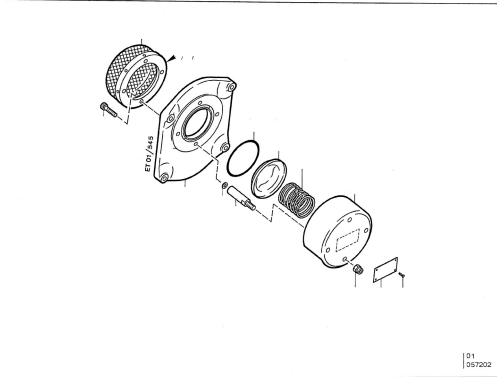
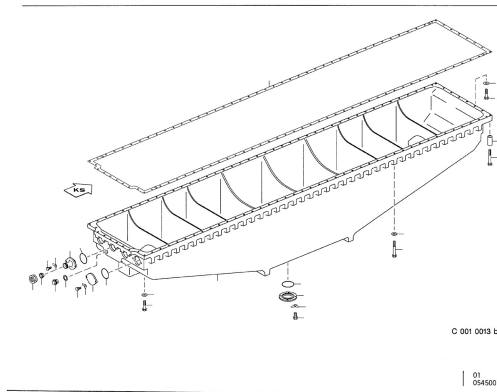
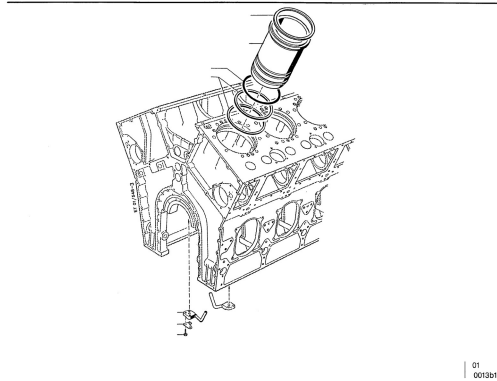
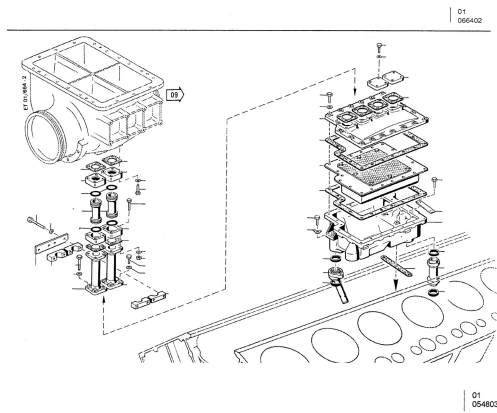
Betriebsstoffe nur nach MTU-Vorschrift No. 1081

Only use fluids and lubricants as per MTU Specification No. 1081.
Utiliser les ingrédients prescrits par Publication MTU No. 1081.
Utilizar solamente materiales de servicio seg. Publ. MTU No. 1081.
Se utilizar agentes de funcionamiento seg. Publ. MTU No. 1081.
Impiegare solo materiali di consumo seg. Publ. MTU No. 1081.

Einbauanleitung

Einbauanleitung





This technical drawing shows an exploded view of a 12-cylinder engine block and cylinder head assembly. The main components are the engine block (labeled 1) and the cylinder head (labeled 2). The diagram illustrates the various bolts and nuts used to secure the head to the block, including the main head bolts (labeled 3), the head gasket (labeled 4), and the head nuts (labeled 5). The intake manifold (labeled 6) is shown attached to the side of the block. The exhaust manifold (labeled 7) is shown attached to the bottom of the block. The diagram also shows the timing belt (labeled 8) and the timing belt cover (labeled 9). The diagram is a black and white line drawing with various labels and dimensions.

mtu
M.A.N. Maybach Mercedes-Benz
Dieselmotor

Typ _____
Motor-Nr. _____
Baujahr _____
Leistung _____
Drehzahl _____ 1/min
Gewicht _____ kg

Motoren- und Turbinen-Union
Friedrichshafen GmbH

34 444 425

1

Hinweise für Betrieb und Wartung

Motor-tyt _____ Ventilspiel bei [°C] _____
Motornummer _____ Einlaß [mm] _____
Startfüllung _____ Auslaß [mm] _____
Einspritz-Förderbeginn [°] _____ Vorwärmtemperatur [°C] min. _____
Düsenöffnungsdruck [bar] _____ Öltemperatur [°C] max. _____
Düsentyp _____ Kühlwassertemperatur [°C] max. _____
Zündfolge _____

Alle weiteren technischen Daten sind der Betriebsanleitung zu entnehmen

34 444 425

7

Warning

Do not open hot crankcase for at least 10 minutes after shutdown of engine.
Do not restart overheated engine before cause of overheating has been remedied.

34 444 425

2

Vor Tausch des Filtereinsatzes unbedingt Ölschlamm ablassen.
Prior to replacement of filter insert oil sludge must positively be drained.
Avant de remplacer l'élément filtrant, il faut absolument vider les crasses d'huile.
Antes de substituir o elemento filtrante, é indispensável despejar o lama de óleo.
Antes de cambiar el elemento filtrante, es imprescindible evacuar el lodo de aceite.
Prima di sostituire la cartuccia filtrante è assolutamente necessario di scaricare la merchia.

ET 01/660-4

4

Betriebsstoffe nur nach MTU-Vorschrift Nr. 1061
Only use Fluids and Lubricants as per MTU Specification No.1061
Utiliser les Ingrédients prescrits par Publication MTU No. 1061
Utilizar solamente materias de servicio seg. Publ. MTU No.1061
Só utilizar agentes de funcionamiento seg. Publ. MTU No. 1061
Impiegare solo materiali di consumo sec. Pubbl. MTU No. 1061

34 444 425

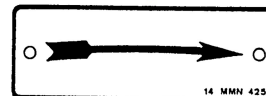
3

Sach-Nr. _____
Baujahr _____
Medium _____
Nenninhalt _____ dm³
Zu-Betriebsüberdruck _____ bar
Zu-Betriebstemperatur _____ °C

mtu

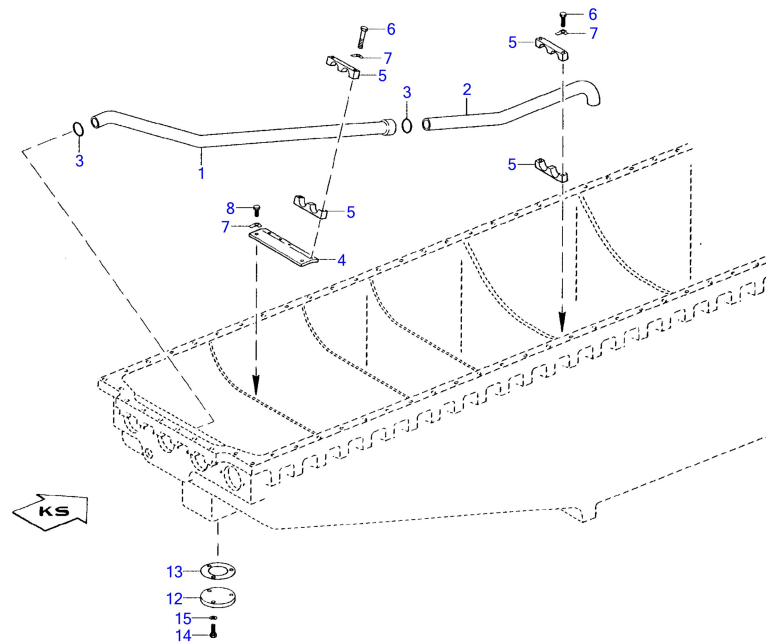
34 444 425

6

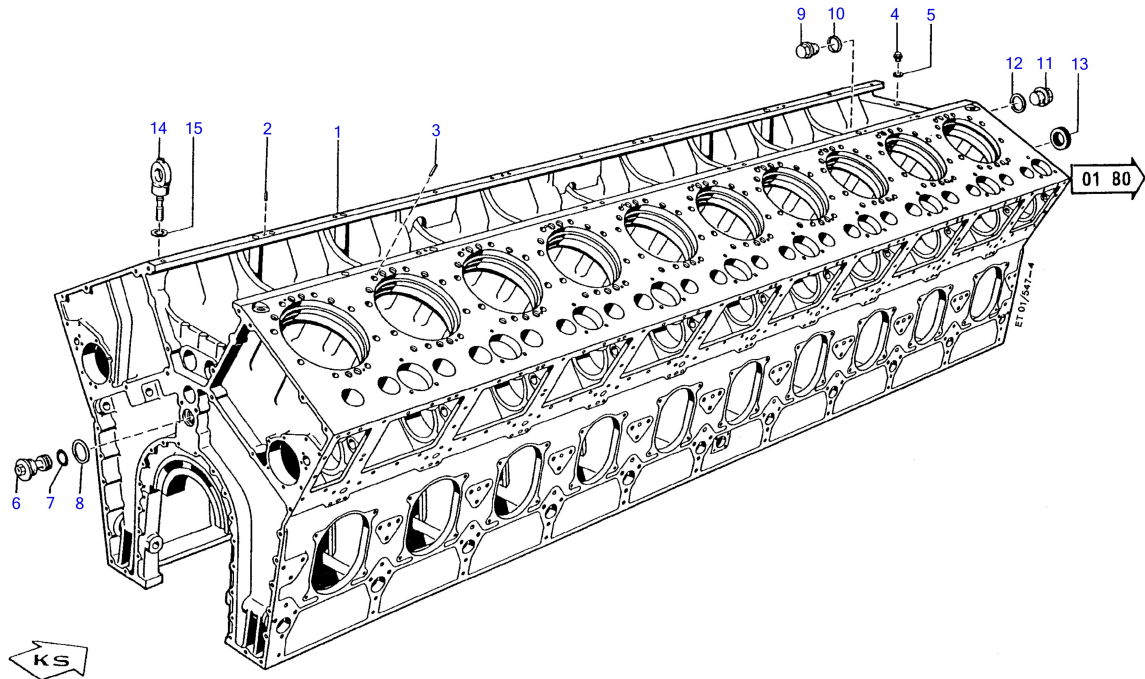


5

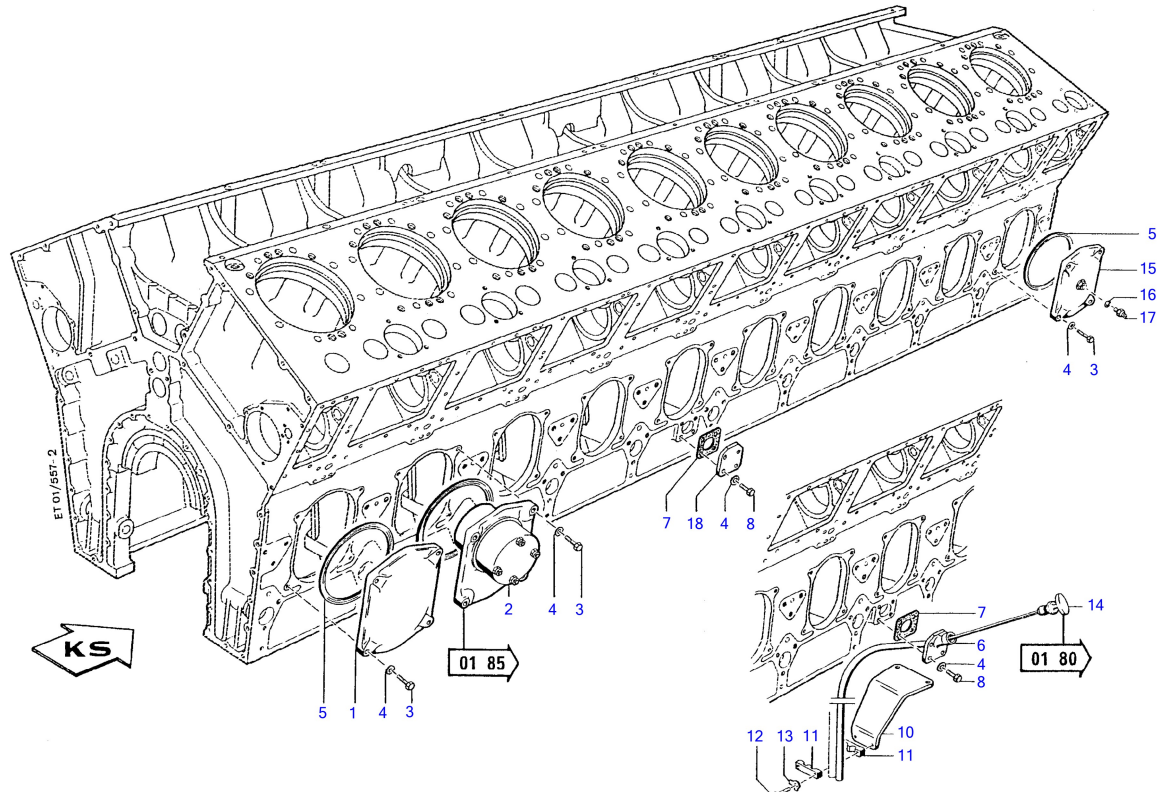
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01	700425000005	INSTRUCTION PLATE	1	EA		04		
02	700425000039	INSTRUCTION PLATE	1	EA		36		
03	700425000108	INSTRUCTION PLATE	1	EA	NB	40.3		
04	700425000109	INSTRUCTION PLATE	1	EA		41.4		
05	700425000025	INSTRUCTION PLATE	1	EA		14		
06	700425000057	INSTRUCTION PLATE	5	EA		42		
07	700425000073	INSTRUCTION PLATE	1	EA	NB	51		



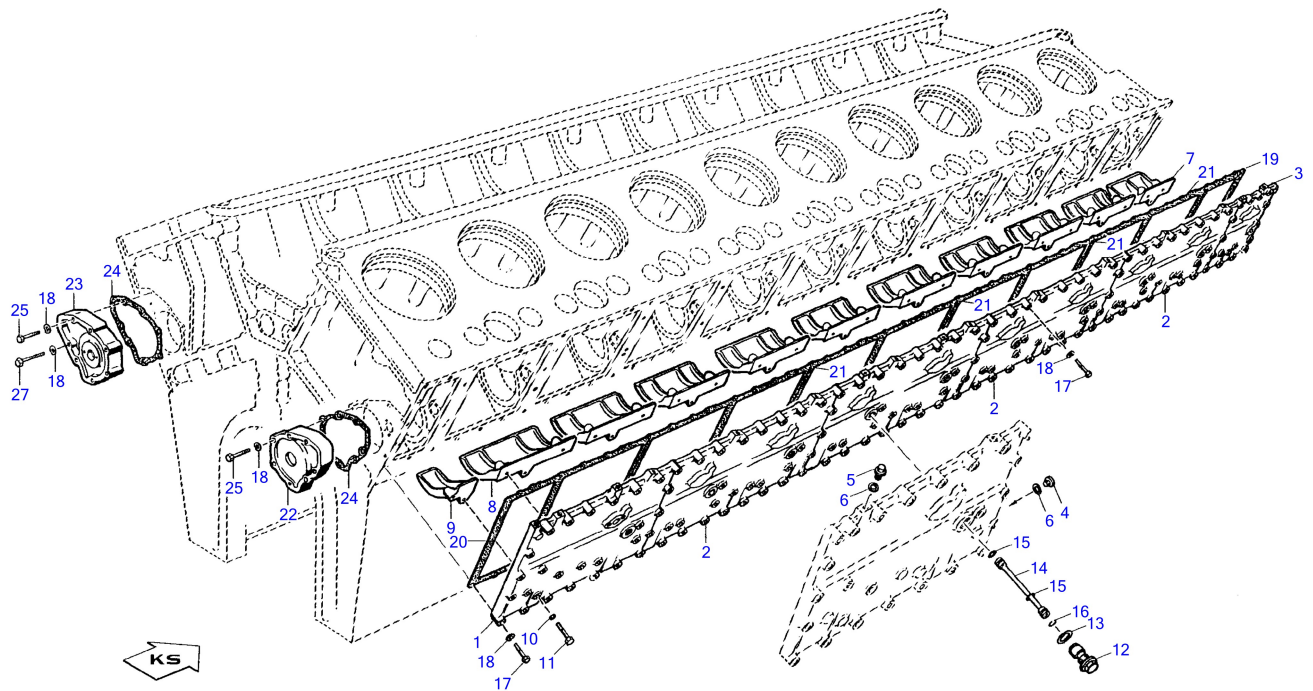
POS.	PART NO.	DESIGNATION	QTY.	UNITS	FN	MEASUREMENT	STANDARD	DES. ADDITION
01	5841800451	SUCTION PIPE	1	EA				
02	5841860125	OIL SUCTION PIPE	1	EA				
03	700429035000	O-RING	1	EA		35,5 X 4		
04	5841860040	ALARM SWITCH	2	EA				
05	5849955068	PIPE HALF-CLAMP	6	EA				
06	000931008227	SCREW	8	EA		M 8 X 55		
07	700292008400	TAB WASHER	14	EA		8,4		
08	000933008108	SCREW	4	EA		M 8 X 20		
12	5900140351	BLIND FLANGE	1	EA				
13	5900141280	GASKET	1	EA				
14	000933010092	SCREW	3	EA		M 10 X 30		
15	000137010202	SPRING WASHER	3	EA		B 10		



POS.	PART NO.	DESIGNATION	QTY.	UNITS	FN	MEASUREMENT	STANDARD	DES. ADDITION
01	5800109008	CRANKCASE	1	EA				KOMPLETT
02	000007008115	DOWEL PIN	4	EA		8 M6 X 18		
03	001481008001	SPRING PIN	40	EA		8 X 20		
04	000910018003	PLUG	1	EA		M 18 X 1,5		
05	007603018101	SEALING RING	1	EA		A 18 X 22		
06	5800110235	PLUG	1	EA				
07	700429048001	O-RING	1	EA		48 X 5		
08	007603064100	SEALING RING	1	EA		A 64 X 72		
09	000910048000	PLUG	1	EA		M 48 X 1,5		
10	007603048101	SEALING RING	1	EA		A 48 X 55		
11	000910052000	PLUG	1	EA		M 52 X 1,5		
12	007603052101	SEALING RING	1	EA		A 52 X 60		
13	5800150560	SEALING RING	2	EA				
14	5802230041	LIFTING EYE	4	EA				
15	5809900240	COVER DISC	4	EA	WW			
15A	5809900940	COVER DISC	4	EA	WW			
15B	5809901040	COVER DISC	4	EA	WW			
15C	5809901340	COVER DISC	4	EA	WW			



POS.	PART NO.	DESIGNATION	QTY.	UNITS	FN	MEASUREMENT	STANDARD	DES. ADDITION
01	5800110432	INSPECTION PORT COVER	15	EA				
02	0000180529	SAFETY VALVE	4	EA	P			
03	000933010093	SCREW	80	EA		M 10 X 35		
04	000137010202	SPRING WASHER	22	EA		B 10		
04Z	000125010524	COVER DISC	66	EA		A 10,5		
05	0029971248	RUBBER RING	20	EA				
06	5800100566	GUIDE TUBE	1	EA	S			
07	5800180580	GASKET	2	EA				
08	000933010035	SCREW	8	EA		M 10 X 25		
10	5800181440	ALARM SWITCH	1	EA				
11	700325018105	PIPE HALF-CLAMP	2	EA		2/2 D 9 - 18		
12	000933008111	SCREW	2	EA		M 8 X 40		
13	700292008400	TAB WASHER	2	EA		8,4		
14	5800102572	OIL DIPSTICK	1	EA				
15	5800100215	INSPECTION PORT COVER	1	EA				
16	007603022102	SEALING RING	1	EA		A 22 X 27		
17	915013016002	EXHAUST PIPE NECK	1	EA		16		
18	5800180544	PRESSURE TEST FLANGE	1	EA				



POS.	PART NO.	DESIGNATION	QTY.	UNITS	FN	MEASUREMENT	STANDARD	DES. ADDITION
01	5800103935	END COVER	2	EA				
01B	5800113110/80	CYLINDER LINER	0	EA	M			STUFE 2
02	5800103635	END COVER	8	EA				
03	5800103835	END COVER	2	EA				
04	000908012003	PLUG	10	EA		M 12 X 1,5		
05	000910012000	PLUG	22	EA		M 12 X 1,5		
06	007603012111	SEALING RING	32	EA		A 12 X 15,5		
07	5840100062	OIL COLLECTING TRAY	2	EA				
08	5840100162	OIL COLLECTING TRAY	18	EA				
09	5840100262	OIL COLLECTING TRAY	2	EA				
10	007603010107	SEALING RING	64	EA		A 10 X 16		
11	000933010150	SCREW	64	EA		M 10 X 55		
12	5809970132	PLUG	20	EA				
13	007603027100	SEALING RING	20	EA		A 27 X 32		
14	5800150155	PISTON OIL DUCT	20	EA				
15	700429010000	O-RING	40	EA		10 X 2		
16	007993014200	SNAP RING	20	EA		B 14		
17	000933010095	SCREW	154	EA		M 10 X 50		
18	000125010524	COVER DISC	168	EA		A 10,5		
19	5800110880	GASKET	2	EA				
20	5800110980	GASKET	2	EA				
21	5800111080	GASKET	8	EA				
22	5800152630	END COVER	1	EA				